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A corpus-based study of semantic prosody in *ràng* and *shǐ* constructions

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Abstract: This study investigates the semantic nuances of the Mandarin causative constructions *ràng* and *shǐ*, focusing on their distinct semantic preferences and variations in semantic prosody. Analysis of 1,000 instances of each construction from the Corpus of Contemporary Taiwanese Mandarin (COCT) Written Corpus 2020 reveals that *ràng* frequently co-occurs with verbs and terms related to intelligence and cognition, while *shǐ* is more often found with terms and verbs associated with psychological states. This suggests that *ràng* constructions are often linked to cognitive processes, while *shǐ* constructions tend to evoke emotional or psychological responses. Although both constructions show an affinity for verbs describing change of state, psychological states, and motion, *shǐ* exhibits greater versatility, appearing with a broader range of verb classes. Examination of semantic prosody, utilizing Stubbs' (1995) classification (positive, negative, neutral), shows that *ràng* constructions exhibit greater variation in their connotational coloring compared to *shǐ* constructions. Notably, both constructions tend to shift towards negative semantic prosody within the "changed" causative events, as shown in Table 6, with *ràng* demonstrating a stronger tendency to negative semantic prosody in comparison with *shǐ* (*ràng*: 47.7 % vs. *shǐ*: 38.9 %). These findings highlight the subtle ways in which *ràng* and *shǐ* constructions convey different shades of meaning. The study contributes valuable insights into the linguistic expression, deepening our understanding of causative constructions in Mandarin.

Keywords: causative construction; collostructional analysis; semantic prosody

提要: 本研究再探漢語致使構式「讓」與「使」的語意細微差異， 聚焦於兩者在語意偏好 (semantic preference) 與語意韻 (semantic prosody) 上的異同。本研究自《現代台灣華語語料庫 COCT》 2020 年書面語料中, 各擷取 1,000 例「讓」構式與 1,000 例「使」構式進行分析。結果顯示, 「讓」構式常與涉及智力與認知的動詞

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共現, 而「使」構式則較常與表達心理狀態的詞語與動詞搭配, 凸顯「讓」構式更傾向於表達認知過程, 「使」構式則較易引發情感或心理層面之反應。雖然兩者皆常與描述狀態變化、心理狀態及移動的動詞共現, 「使」構式在動詞類別上的搭配範圍更為廣泛。我們進一步依據 Stubbs (1995) 之分類 (正面、負面、中性) 檢視語意韻, 發現「讓」構式在語意色彩上變化較大; 值得注意的是, 在語意韻發生變化的致使事件中, 兩種構式雖皆呈現向負面語意韻的轉移, 而「讓」構式在負面語意韻變化相較於「使」構式更明顯 (讓: 47.7 % vs. 使: 38.9 %)。本研究凸顯「讓」與「使」在語言表達上的細微差異, 深化我們對漢語致使構式的理解。

關鍵詞: 致使構式; 構式搭配分析; 語意韻

1 Introduction

The study delves into the semantic complexity and grammatical functionalities encapsulated within the Mandarin causative constructions, specifically *ràng* and *shǐ*. The two polysemous lexicons construct periphrastic causations, a phenomenon that has captured the sustained interest of scholars in the fields of Mandarin linguistics and pedagogy studies. This study interrogates two primary concerns: firstly, the discernible differences in verb lexeme distribution when *ràng* and *shǐ* serve as the nucleus of the construction; and secondly, the distributional contrasts of their overlapping collocate across both constructions. The application of construction grammar reveals that constructions serve as the fundamental linguistic units where semantics and grammatical functionalities converge (Goldberg 1995). This theoretical framework is instrumental in unraveling the multifaceted senses of *ràng* and *shǐ* – including passive, causative, permissive, and concessive interpretations – as well as the dynamic syntactic interactions, such as those exemplified by the ditransitive construction (Weng 2007). These elements collectively constitute the complexities and challenges that researchers and learners encounter when exploring the causative verbs *ràng* and *shǐ* in Mandarin.

Extensive quantitative research has been conducted on *ràng* and *shǐ*, employing diverse methodologies from behavioral profile analysis to logistic regression models and collostructional analysis. Studies have investigated the distributional variances of these causative constructions across multiple linguistic registers, verb semantic prototypes, and regional dialects (Liu and Li 2023; Tian et al. 2022; Yang 2016). Liu and Li (2023) concluded that *ràng* construction attracts psychological verbs and verbs of getting information, while *shǐ* construction attracts verbs of change-of-state and change-of-possession. *Shǐ* construction is more often used in written language, while *ràng* construction is more often used in spoken language. In addition, *shǐ* construction

is often used in Mainland China, while *ràng* construction is more often used in Taiwan (Tian et al. 2022). Despite the comprehensive claims provided by earlier studies and acknowledging that causative events pivot around the verbal lexeme central to the construction (NP_1 +*shǐ*/*ràng* + NP_2 + VP) highlighted by Gilquin (2006), there remains unclear research concerning the influence of causative verbs such as *ràng* and *shǐ*, on the overall semantic prosody (Sinclair 1991) within causative constructions in Mandarin Chinese. The research questions are mentioned below:

1. What are the semantic preferences of the verb collexeme within the causative constructions of *ràng* and *shǐ*?
2. Does the semantic prosody shifting from left context of causative verbs to the right prefer a different tendency between the *ràng* and *shǐ* causative constructions?

2 Literature review and theoretical framework

Goldberg (1995) introduced the theory of construction grammar, defining a 'construction' as the smallest unit of meaning, embodying a complex relationship between semantics and grammatical functions. This theoretical framework facilitates a more comprehensive analysis of causative events by allowing researchers to take a look at the connections of the meaning and form in a more comprehensive way. Mandarin causative events, characterized by periphrastic causation (Kemmer and Verhagen 1994), exhibit complexity in their causative constructions, particularly with the polysemous nature and grammatical interactions of causative verbs *ràng* and *shǐ* (Weng 2007). Zhang (2005, 2006) investigated the development of Mandarin causative events which show the semantic complexity of causative verbs. Chinese causatives usage revolutionize from permissive to causative which make it unclear to distinguish causative senses from the polysemes, with such evolution process verified by implementing quantitative methods on the micro-relatives of causative construction family (Chen 2025). Wang (2011) conducted a semantic, discourse, and pragmatic analysis on causative polysemy *ràng*. *Ràng*$_1$ obtains abstract and continuous cause-effect event usages, and *ràng*$_2$ obtains permissive meaning. These two usages of *ràng* show differences in parts of speech, syntax, and discourse level, while *ràng*$_2$ has a higher usage frequency in Mainland China rather than Taiwan. Weng (2007) revealed the unique features of Mandarin's periphrastic causative structures, highlighting the senses of *ràng* as causative, permissive, and passive, and the grammatical functions derived from the interaction between causative verbs with syntax structures, presenting significant challenges for quantitative research in Mandarin causation. The research above shows the necessity of analyzing causative verbs and syntax structures as a complete aspect.

Recent advances in construction grammar, increasingly convergent with the usage-based outlook of cognitive grammar, provide theoretical foundation and quantitative methods for the present study. Boas et al. (2024) show that constructionist frameworks share a view of form–meaning pairings and recognize frequency-sensitive generalizations. Diachronic and corpus-driven work employing distributional vector-space models (Desagulier 2022) maps shifts within construction networks and demonstrates that semantic prosody can be measured through the type of variation examined here for *ràng* and *shǐ*. Computational implementations such as fluid construction grammar (Beuls and Van Eecke 2023) provide explicit procedures for estimating collostructional strength in usage data, complementing the Fisher-exact tests used in this article. Cognitive construction grammar sharpens this picture by insisting that all linguistic knowledge consists of learned pairings of form and meaning, from morpheme to discourse pattern (Boas 2013). It further argues that the organization of those pairings is usage-based, frequency-sensitive, and motivated by general cognitive principles such as iconicity and prototype categorization. Within the framework, constructions populate taxonomic networks whose inheritance links capture both schematic regularities and item-specific exceptions – an architecture that mirrors the gradient shifts we observe in the collostructional profiles of Mandarin causatives. These developments justify the combination of collostructional and semantic prosody approach adopted here to position Mandarin causative constructions.

Collostructional analysis, initially introduced by Gries and Stefanowitsch (2004), offers a nuanced methodology for exploring the semantic and syntactic compatibility between lexical items and grammatical constructions. It builds upon the foundational principles of construction grammar (Goldberg 1995), which treats grammatical constructions as the minimal units of meaning, integrating form and function. The method identifies collocational preferences within specific grammatical slots, revealing subtle distinctions that traditional corpus analysis might overlook. Three primary types of collostructional analyses have been developed to address varying linguistic phenomena: (1) simple collexeme analysis, which evaluates the strength of association between specific lexical items and a particular grammatical construction; (2) distinctive collexeme analysis, comparing lexical preferences across competing constructions to discern fine-grained distributional differences; and (3) covarying collexeme analysis, examining interactions between lexemes within complex constructions to uncover syntactic or semantic correlations. These approaches are particularly valuable for analyzing alternations, such as the dative alternation or causative constructions, providing insights into the cognitive and pragmatic factors influencing linguistic choices (Gries and Stefanowitsch 2003). In applying these methods, studies of Mandarin causatives (e.g., *ràng* and *shǐ*) have illuminated

distinctions in semantic roles and prosodic tendencies, advancing our understanding of linguistic variability and typological universals.

Quantitative research in Mandarin causation has been extensive regarding implemented methods such as covering linguistic variables, semantic frames, and collocations. Yang (2016) explored the differences in verb collocations, registering categories in *shǐ*, *jiào*, and *ràng* using collostructional analysis, and finding out that psychological verbs are more distinctive with *ràng* constructions. *Ràng* construction is more frequently used for causative events in Taiwan, and *shǐ* construction is more frequently used for causative events in Mainland China. Liesenfeld et al. (2022) investigated the differences between *shǐ*, *ràng*, and *lìng* using behavioral profile (BP) methods which highlights the importance of implementing semantic frame elements (FE) (Fillmore 2006) into quantitative polysemy analysis. It notes the significant differences between *lìng* and *shǐ* and *ràng*, while *shǐ* and *ràng* show their differences on the quantitative distribution of semantic frame element. Shih et al. (2020) employed multiple regression models to examine the differences in two dimensions of animacy and physicality of first noun phrase position of the causative construction of *shǐ* and *ràng*, determining their roles in direct and indirect causative events. Tian et al. (2022) used random forest tests to ascertain the effects of several variables on the distribution of Mandarin causatives, confirming the impacts of linguistic variation and the distinction between written and spoken language. Liu and Li (2023) analyzed the verb collocations of the VP_2 slot in *ràng* and *shǐ*, investigating the prototypical semantics of such. Most recently, Chen (2025) investigated the construction types, lexical roles, and contextual factors of Chinese analytic causative construction, revealing how different micro-constructions evolve and redistribute their semantic functions over time. These studies show the multidimensional quantitative evidence of Mandarin causation differences in verb lexeme, prototypical meaning, distribution differences in regional variants, and spoken and written context, however the inner fine-grained semantic variability of the causative construction is still left unclear.

Despite these studies sketching a profile of Mandarin causative constructions, the internal semantic changes within causative events remain somewhat vague (e.g., whether the meaning changes during the causative event). Sinclair (1991) proposed the concept of 'semantic prosody' as "the spreading of connotational coloring beyond single word bound." The third layer of Partington's (1998) definition of 'connotational coloring' is an expressive connotation which infers that the chosen words would reflect the different attitude of the speaker toward the object. Therefore, this study focuses on the verb lexeme as the core, adopting Stubbs' (1995) typology of positive, negative, and neutral semantic prosody to observe the contexts collocated with the construction from the left to the right (e.g.: In the construction: *Yōuměi de shèyǐng jí yīnyuè de chèntuō xià, gèng ràng rén juédé ài bùkě dé de gǎnshāng* 'Beautiful photography and music make people feel the sadness of love being unavailable', the

semantic prosody shifts from positive to negative from the left to the right side of the construction). This approach aims to provide a more comprehensive understanding of internal semantic variations within Mandarin causative constructions. The semantic variations appeared based on the dependence between semantic prosody and syntactic structure (Lin and Chung 2016). Bednarek (2008) distinguished semantic prosody and semantic preference as follows. Semantic prosody is defined as the collocations with certain connotational coloring, and is categorized into negative, positive, and neutral. As for semantic preference, it is defined as the tendency for lexical items to be restricted to identifiable semantic fields (Cheng 2011) and viewed as a semantic environment in which vocabulary items commonly occur (Phoocharoensil 2021). Building on Bednarek's definition, the study adopts semantic preference to explain the distribution of 15 verb classes and semantic prosody to tag the connotational coloring shifts of the two endpoints of both constructions.

Despite the considerable body of research on Mandarin causatives, critical gaps remain in understanding the contextual and pragmatic nuances of the causative verbs *shǐ* and *ràng*. While prior studies have distinguished direct causation, typically expressed by *shǐ*, from indirect causation associated with *ràng* (Liu and Li 2023; Verhagen and Kemmer 1997), further investigation is needed into how social and discourse contexts might influence verb selection. Factors such as politeness strategies and speaker intention may play a key role in this process. Additionally, while collocational tendencies have been examined – with *shǐ* often paired with verbs of physical or psychological change and *ràng* with verbs related to perception and cognition (Liesenfeld et al. 2022; Yang 2016) – the semantic polarity (positive, negative, or neutral) of these constructions remains underexplored. This gap highlights the need for further detailed studies on how contextual factors shape polarity patterns.

To sum up, we address one of these research gaps, aiming at contributing to a better understanding of Mandarin causatives, bridging the existing theoretical and practical knowledge on their syntax, semantics, and pragmatic usage.

3 Methodology

3.1 Data source

This study extracted data from the Corpus of Contemporary Taiwanese Mandarin (COCT), specifically the Written Corpus 2020.[1] Unlike other studies that might screen

1 The present study utilizes the Written Corpus 2020, a sub-corpus of the Corpus of Contemporary Taiwanese Mandarin (COCT), compiled and released by the National Academy for Educational

corpora directly by semantic categories, this research employs the CQP syntax filtering tool to select and retrieve corpus data related to *ràng* and *shǐ* construction. To capture sentence-level tokens of the two causative verbs, we employed the following CQP patterns (CKIP POS tags in square brackets; wildcards allow for intervening function words):

// *ràng* construction

[word="讓"][pos="Na"][pos="VA" | pos="VB" | pos="VC" | pos="VD" | pos="VE" | pos="VF" | pos="VG" | pos="VH" | pos="VHC" | pos="VI" | pos="VJ" | pos="VK" | pos="VL"]

// *shǐ* construction

[word="使"][pos="Na"][pos="VA" | pos="VB" | pos="VC" | pos="VD" | pos="VE" | pos="VF" | pos="VG" | pos="VH" | pos="VHC" | pos="VI" | pos="VJ" | pos="VK" | pos="VL"]

Initial filtering using CQP syntax excludes grammatical functions that interact with causative verbs[2] (for example, filtering out structures with *le* as in *wǒ* ràng *le tā yī-gōngchǐ* – 'I gave him one meter'). Subsequent manual refinement excludes passive, permissive, and concessive sentences, leaving 1,000 instances[3] each for the *ràng* and shǐ constructions.

3.2 Collostructional analysis in semantic preferences

Semantic preference classification requires further quantitative validation. To achieve this, we investigated the collocation data with the constructions statistically. Namely, we focus on *ràng* and *shǐ* as Mandarin causative constructions, with the verb slot in the canonical structure $N_{Causer} + V_{Cause} + N_{Causee} + V_{Affectee}$ (Xiong and Liang 2003) being subjected to a family of collostructional analyses (Gries and Stefanowitsch 2003; Hsiao and Lestari 2020). The first phase utilizes simple collexeme analysis, which determines the semantic compatibility of collexemes with the respective constructions. This involves calculating the collocational strength of verbs within the *ràng* and *shǐ* constructions to discern their alignment with these causative

Research (NAER) in Taiwan. This sub-corpus contains approximately 11.3 million tokens drawn from nine written genres – including news reports, academic prose, and online essays – produced between January and December 2020.

2 Given the nuanced tagging system of the COCT corpus, the authors opted to include all potential verb-related results retrieved from the query.

3 A total of 2000 instances were selected following the use of the corpus's 'shown in random order' function. Duplicate entries were removed, and the first 2000 unique results from the query were retained.

Table 1: Co-occurrence table for a simple collexeme analysis.

	CXC_x	$Oth.C_x$	R.Total
Word Wx	a	b	a + b
Other W	c	d	c + d
Col.Total	a + c	b + d	N = a + b + c + d

Note: CXCx refers to target construction frequency in corpus; Other refers to other construction frequency in corpus; R.Total refers to row total; Col.Total refers to column total.

frameworks (Table 1). The co-occurrence table summarizes the frequency relationships as follows:

In the second phase, distinctive collexeme analysis is employed (Table 2), which examines overlapping collexemes between the two constructions. This approach assesses whether a collexeme exhibits a stronger attraction to *ràng* or *shǐ* constructions. Fisher's exact test is used to determine the statistical significance of collocate co-occurrences, ensuring that semantic proximity is not merely a product of high-frequency patterns but instead reflects a substantive relationship with the constructions. The results yield the top 20 significant collocations for both analyses. These collocations are then classified into Levin's (1993) 15 verb classes, elucidating the semantic core of each construction. Subsequently, the analysis quantifies the collocational strength of overlapping collexemes, highlighting distinctions between the two constructions. The attracted collexemes of each construction are further categorized into verb classes to identify significant semantic core differences. The statistical calculations for both analyses are conducted using the R statistical software (R Core Team 2023), specifically leveraging the {coll.analysis 3.2a} package (Gries and Stefanowitsch 2004).

3.3 Semantic prosody analysis

To examine the internal shifts in semantic prosody within the constructions, we analyzed changes in prosodic alignment from the left to the right side of the clause. From the dataset of 1,000 instances each of *ràng* and *shǐ* constructions, 500

Table 2: Co-occurrence table for a distinctive collexeme analysis.

	CXC_1	CXC_2	R.Total
Word Wx	a	b	a + b
Other W	c	d	c + d
Col.Total	a + c	b + d	N=a+b + c + d

representative examples from each group were extracted for detailed analysis. Concordance indexing was employed to trace prosodic changes by inspecting the surrounding lexical context and observing paragraph-level shifts in meaning. Semantic prosody in the experimental data was annotated and analyzed based on Stubbs' (1995) tripartite classification: positive, negative, and neutral. The study identified various patterns of prosodic shifts across the constructions. For example, in the sentence *Zhèngyào de hūnyīn shēnghuó yě bù měimǎn, ràng shìqíng biàn dé gèng fùzá* 'The marriage life of political figures is also unhappy, which makes things more complicated', the prosody remains neutral throughout and is thus categorized as a no-change case. The analysis further investigates whether systematic differences exist in the directionality (from left to right) of semantic prosody shifts between the two constructions. Preliminary observations suggest that *ràng* is generally associated with negative changes, whereas *shǐ* shows a greater tendency toward neutral shifts. The distributional differences in semantic prosody between *ràng* and *shǐ* constructions are then summarized and discussed.

4 Results and discussion

This study examines the semantic features and variability of verb collexemes in Mandarin causative constructions of *ràng* and *shǐ*, offering a detailed exploration of their distinct roles in framing causation. Simple collexeme analysis highlights unique collocational patterns and semantic traits, with the "Coll: strength" index (Tables 3 and 4) revealing robust collexeme-construction associations. For example, four-character expressions like *huīxīn-yīxuàn* 'relieved' and *pāida-jiàojué* 'clapping hands' dominate *ràng*, reflecting its preference for external, observable changes. In contrast, emotional terms like *máogǔ-sǒngrán* 'fearful' and *hánxīn* 'uncertain' align strongly with *shǐ*, underscoring its psychological and introspective focus.

Ràng emphasizes dynamic external transformations through verbs such as *chōngxiàpō* 'running downhill', which demonstrate its alignment with physical and situational shifts. *Shǐ*, however, reflects internal causation, favoring verbs like *xièqì* 'to discourage' (indicative of psychological processes). This divergence, as shown in Tables 3 and 4, extends to motion verbs: *pāida-jiàojué* 'clapping hands' aligns with *ràng*'s emphasis on tangible actions, while *luòlèi* 'shedding tears' corresponds to *shǐ*'s emotional and cognitive resonance. Further distinctions emerge in the psychological depth captured by *shǐ*'s collexemes, such as *zùoérwángyòu* 'completely lost', which reflect nuanced internal states. *Ràng*, meanwhile, gravitates toward tangible contexts with verbs like *dākāi-yǎnjiè* 'to broaden', emphasizing its role in fostering visible and actionable outcomes. These findings deepen our understanding of how

Table 3: The top 15 distinctive collexeme found in *ràng* construction.

Collexeme	Corpus. freq	Constr. freq	Coll.strength
huìxīn-yīxiào	336	11	10.6433957
pāida-jiàojué	2	2	5.55344661
shāojiāodào	1	1	2.77672282
xúnxiàn	1	1	2.77672282
zìlǜ-zìzhòng	1	1	2.77672282
yǎnhuā-liáoluàn	597	5	2.44426911
zhuōmōbùdìng	242	3	2.09120576
bùnài-qífán	6	1	2.00038683
dàkāi-yǎnjiè	510	4	1.95175999
xīndàng-shényí	7	1	1.93380279
díchén	8	1	1.87617349
pěngfù	110	2	1.82757968
yìnxiàng-shēnkè	1,195	6	1.78491277
chōngxiàpō	10	1	1.77998847
sànchūqù	11	1	1.73895813
chóngzhǎn	11	1	1.73895813

ràng and *shǐ* encode causation, illustrating *ràng*'s affinity for externalized change and *shǐ's* broader cognitive and affective dimensions. This nuanced analysis enriches the study of Mandarin causative structures by showcasing their interplay between semantics, syntax, and pragmatics.

Figure 1 shows a similar semantic preference of the verb classes mentioned above: the *shǐ* construction exhibits a broader diversity in the distribution of verb classes (within 15 verb classes, *ràng* involves 4 verb classes at 0 %, whereas *shǐ* has 2 verb classes at 0 %) which indicates that *shǐ* construction accommodates a broader semantic domain. The distinct semantic tendencies underscore *ràng's* inclination toward framing external changes and actions, while *shǐ* integrates more internalized and psychological dimensions, offering broader applicability across verb classes. Figure 1 further delineates these patterns, showing that while *ràng* excludes four verb classes entirely, *shǐ* excludes only two, underlining its greater semantic versatility and breadth.

These refined insights not only deepen our understanding of Mandarin causative constructions but also enrich broader discussions in cognitive linguistics and construction grammar by revealing how these constructions encode meaning through collocational preferences and prosodic shifts.

Distinctive collexeme analysis highlights how overlapping collexemes tend to be associated with one of the constructions. As shown in Table 5, the *p-value* confirms

Table 4: The top 15 distinctive collexeme found in *shǐ* construction.

Collexeme	Corpus. freq	Constr. freq	Coll.strength
máogǔsǒngrán	1,056	10	8.96862002
qǐjìng	56	4	7.40105818
èwàn	353	6	7.09712286
xīnfú	236	5	6.47587058
yǎnhuā-liàoluàn	6	2	5.30239518
hánxīn	217	4	5.04470636
zhànlì	707	5	4.17868092
dèngyǎn	449	4	3.8215161
xièqì	881	5	3.73560738
duòjiǎo	516	4	3.59245063
hàotàn	41	2	3.57051365
tàn-wéi-guān-zhǐ	562	4	3.45281342
luòlèi	1,092	5	3.31185782
dǒnglǐ	1	1	3.23890772
zuòérwàngyōu	1	1	3.23890772

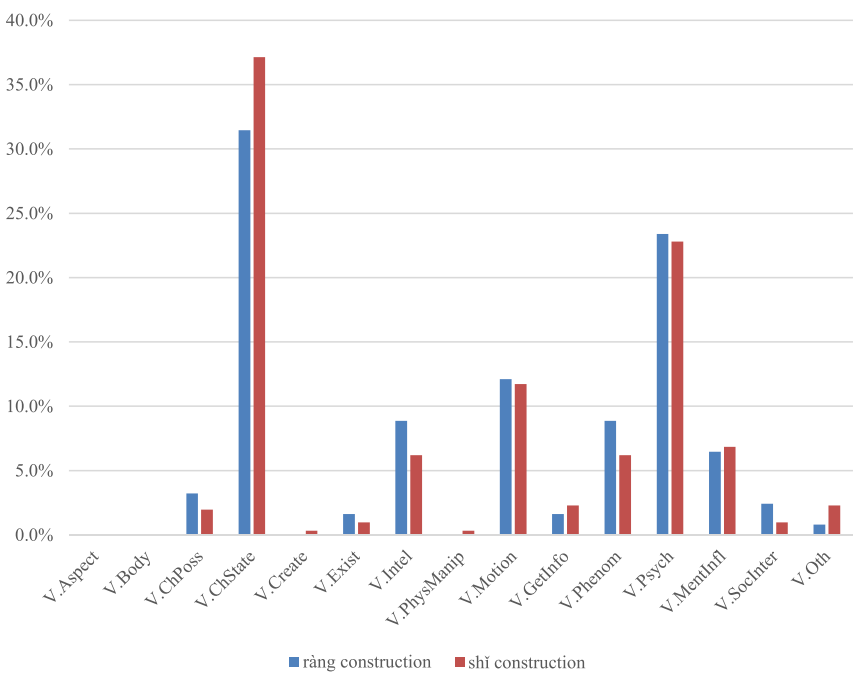

Figure 1: The verb classes of the significant collocations in *ràng* and *shǐ* constructions.

Table 5: The most distinctive 11 collexeme for *ràng* and *shǐ* construction.

***ràng* Construction**		***shǐ* Construction**	
Collexeme	**Coll.strength**	**Collexeme**	**Coll.strength**
yǐwéi	2.72E-05	*biàn*	−0.0001294
gǎnshòudào	2.51E-05	*biànchéng*	−8.19E-05
huìxīn-yīxiào	1.82E-05	*jǐnzhāng*	−5.63E-05
zhīdào	1.95E-05	*máogǔsǒngrán*	−6.06E-05
kāixīn	1.14E-05	*gǎndào*	−0.0001336
xuéhuì	9.20E-06	*dānxīn*	−4.98E-05
liǎojiě	9.20E-06	*jǔsàng*	−3.68E-05
juéde	1.47E-05	*xiǎngdào*	−4.51E-05
xiǎngqǐ	6.96E-06	*xīnfú*	−3.03E-05
kàn	7.42E-06	*róngyì*	−3.03E-05
shòubùliǎo	4.72E-06	*chī jīng*	−3.23E-05
zhǎodào	2.47E-06	*xiǎngqǐ*	−3.86E-05
fāxiào	2.47E-06	*huó*	−2.58E-05
tǐhuì	4.55E-07	*chéngwéi*	−4.22E-05
fāxiàn	2.28E-07	*máogǔsǒngrán*	−2.15E-05

the strength of the collexeme's relation to the *ràng* construction, with higher values indicating stronger associations. *Ràng* favors intelligence and cognition verbs like *yǐwéi* 'think', *zhīdào* 'know', and *xuéhuì* 'learn' (V: Intel, *ràng*: 42.1 %), while *shǐ* prefers psychological state verbs such as *jǐnzhāng* 'nervous', *dānxīn* 'worried', and *jǔsàng* 'upset' (V: Psych, *shǐ*: 39.4 %).

This distinction in verb classes reflects the semantic differences between the constructions, as shown in Figure 2, which reveals the verb classes of overlapping collexemes and their associations with each construction. *Ràng* emphasizes externalized, cognitive actions, while *shǐ* captures a broader spectrum of internal and affective dimensions, highlighting their functional divergence in causative constructions.

A chi-square test of independence compared the frequencies of seven prosody patterns – six 'change' types and one 'no-change' type – across *ràng* and *shǐ* constructions (see Table 6). The association was significant, χ^2 (6, N=1,000) = 52.68, $p < 0.001$, with a small-to-medium effect (*Cramer's V* = 0.23; Cohen 1988). Within the change set, the $N_2 \rightarrow N_1$ shift was markedly more common in *ràng* (15.6 %) than in *shǐ* (6.2 %), $\varphi = 0.15$, $p < 0.001$, and yielded the largest effect among the six 'change' patterns. The $P \rightarrow N_1$ and $N_2 \rightarrow P$ pattern was likewise more frequent in *ràng*, although their effect sizes were smaller ($\varphi = 0.09$ and 0.08, respectively).

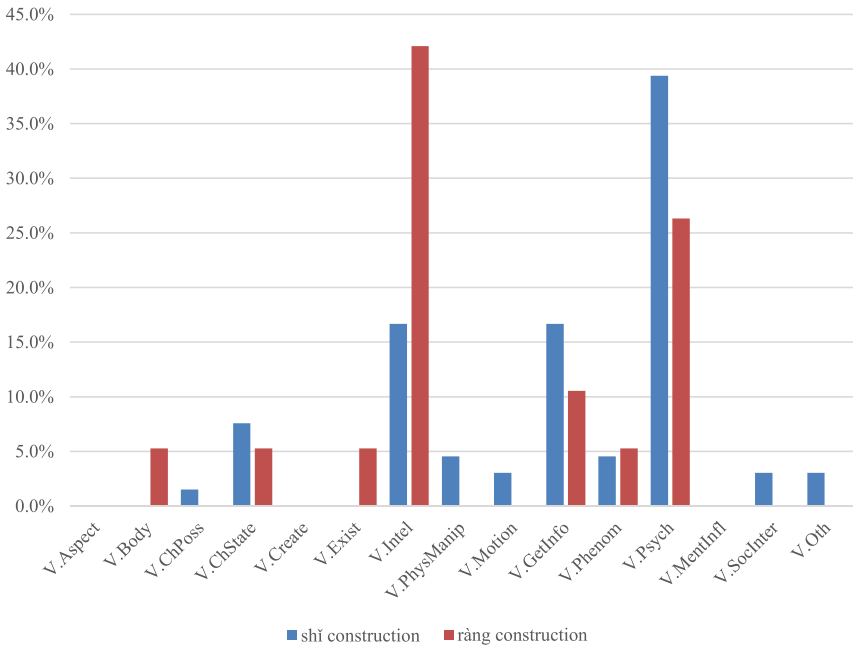

Figure 2: The overlapping verb class and significant collocations in two constructions.

Table 6: The semantic prosody variability.

		***ràng* Construction**	**Total # (percentage)**		***shǐ* Construction**	**Total # (percentage)**	
Change	P-N_1	5(1 %)	83	174	18(3.6 %)	49	126
	N_2-N_1	78(15.6 %)	(47.7 %)	(34.8 %)	31(6.2 %)	(38.9 %)	(25.2 %)
	P-N_2	18(3.6 %)	39		14(2.8 %)	34	
	N_1-N_2	21(4.2 %)	(22.4 %)		20(4 %)	(34.1 %)	
	N_2-P	48(9.6 %)	52		28(5.6 %)	53	
	N_1-P	4(0.8 %)	(29.8 %)		15(3 %)	(27 %)	
No change	P-P	43(8.6 %)		326(65.2 %)	68(13.6 %)		374(74.8 %)
	N_1-N_1	78(15.6 %)			110(22 %)		
	N_2-N_2	226(45.2 %)			195(39.0 %)		

These findings corroborate earlier observations that *ràng* constructions more often shift the original evaluation stance toward the causee (NP_2), bringing it closer to that of the causer (NP_1), whereas *shǐ* constructions tend to preserve the original semantic prosody. Descriptively, Table 6 shows that 34.8 % of *ràng* tokens – but only

25.2 % of *shǐ* tokens – underwent a prosodic shift. For completeness, the three 'no-change' subtypes (P-P, N_1-N_1, N_2-N_2) together accounted for 65.2 % of *ràng* tokens and 74.8 % of *shǐ* tokens, highlighting the comparatively conservative prosodic behavior of *shǐ* constructions.

Figures 3 and 4 illustrate the proportion of 'no change' and 'change' occurrences for both constructions, in order to show not only the differences in the stability of semantic prosody between *shǐ* and *ràng*, but also the shift tendency of semantic prosody towards positive, negative, and neutral from the left to the right of the two constructions. S*hǐ* constructions maintain a higher stability in context, with 74.8 % of instances showing no change in semantic prosody, compared to 65.2 % for *ràng*. Notably, *ràng* displays a larger proportion of changes involving negative or neutral prosody, with higher frequencies of shifts toward P (positive) and N_1 (negative). This highlights the flexibility of *ràng* in expressing evaluative or subjective contexts. Figure 5 compares three primary change patterns (N_1 & $N_2 \rightarrow$ P, N_1 & P $\rightarrow N_2$, N_2 & P $\rightarrow N_1$) between *ràng* and *shǐ* constructions. Across all patterns, *ràng* consistently exhibits a higher frequency of shifts, particularly in the N_2 & P $\rightarrow N_1$ category (47.7 occurrences in *ràng* vs. 38.9 in *shǐ*). These results reinforce the view that *ràng* is more prone to changes involving negative evaluations or subjective reinterpretations of the causee, whereas *shǐ* tends to convey more direct, stable causation with less variability in semantic prosody.

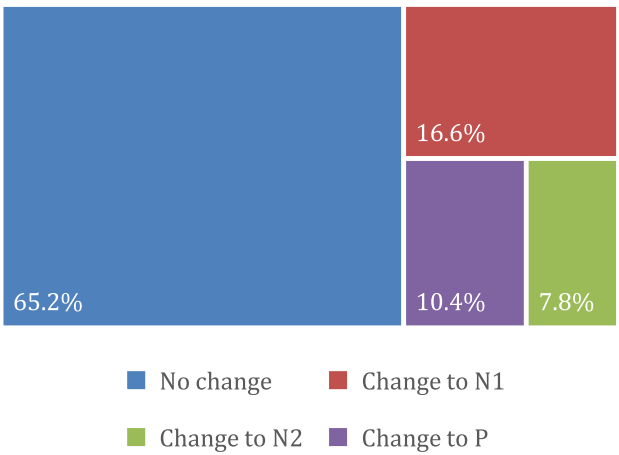

Figure 3: The 'no change' and the sub-categories of 'change' (Note: The numbers in the 'change' sub-categories in Figure 3 and 4 is the sum of the numbers in the columns of Table 6. For example: The number of 'Change to N_1' (16.6 %) is the sum of 'P-N_1' (1 %) and 'N_2-N_1' (15.6 %)) (change to N_1, N_2, P) proportion of *ràng* construction.

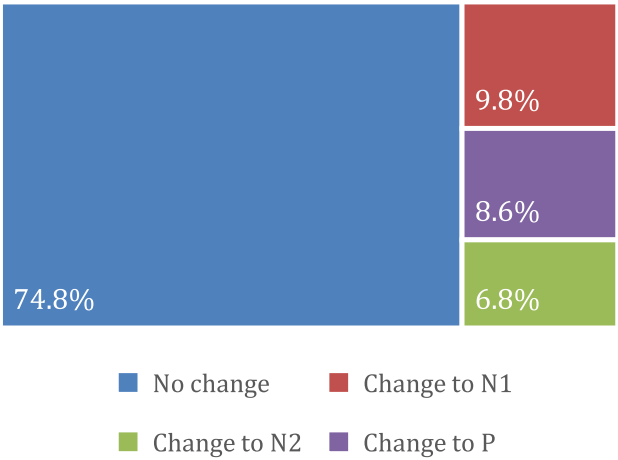

Figure 4: The 'no change' and the sub-categories of 'change' (change to N_1, N_2, P) proportion of *shǐ* construction.

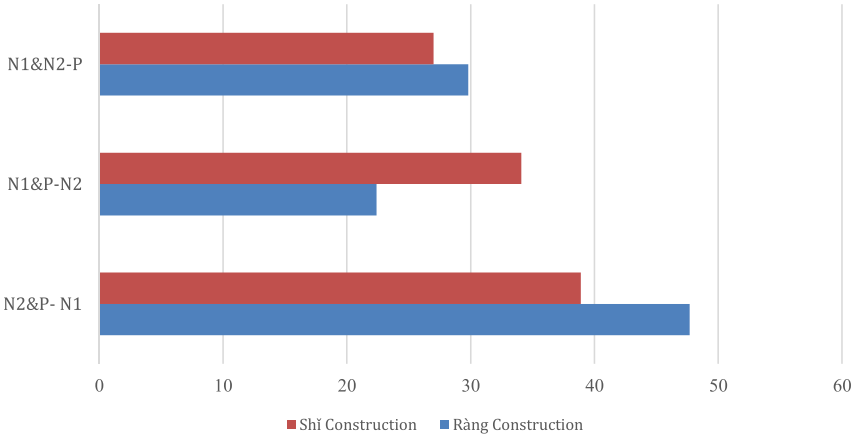

Figure 5: The 'change' proportion of the three categories (N_1 and N_2-P/N_1 and P-N_2/N_2 and P-N_1) two constructions.

The analysis demonstrates that the verb collexemes shared by *ràng* and *shǐ* tend to cluster with similar semantic domains; however, the contrasting attraction and repulsion patterns among their overlapping collocates indicate distinct functional specializations. Specifically, *ràng* is more closely associated with cognition-oriented causative events, while *shǐ* is more aligned with psychologically oriented ones. This distributional profile directly informs the second research question concerning

semantic prosody across clause positions. As the clause progresses from left to right, *ràng* exhibits significantly greater variability than *shǐ*. In causative contexts, both constructions show a tendency to shift toward a negative evaluative stance, with the shift being notably more pronounced in *ràng* than in *shǐ* constructions.

This study advances the understanding of Mandarin causative constructions with *ràng* and *shǐ*, building on and distinguishing from prior research through a fine-grained analysis of semantic prosody changes. In comparison with the earlier collostructional analysis like Yang (2016) and Liu and Li (2023), the distribution of collexemes of this study is quite different, especially with semantic preferences of verb classes. We infer that such a result may be due to the adopted CQP query method which is a strict method that excludes causative events located too far away from *ràng* and *shǐ* in the clause. By analyzing semantic prosody, we identified the preferred context of *ràng* to convey negative semantic prosody more distinctly than *shǐ*. In order to gain deeper insight into semantic preferences and semantic prosody, we plan to expand the comparison to include other causative markers, such as *jiào* and *lìng*, as well as hierarchical and associative causative constructions (Chen 2025; Pijpops et al. 2021; Tian 2025). The study's unique contribution lies in its detailed examination of semantic prosody within the context of verb collocations, offering novel insights into the interplay between cognitive processing and language expression, informing both theoretical linguistics and practical language teaching.

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References

Bednarek, Monika. 2008. Semantic preference and semantic prosody re-examined. *Corpus Linguistics and Linguistic Theory* 4(2). 119–139.

Beuls, Katrien & Paul Van Eecke. 2023. Fluid construction grammar: State of the art and future outlook. In Claire Bonial & Harish Tayyar Madabushi (eds.), *Proceedings of the First International Workshop on Construction Grammars and NLP (CxGs+NLP, GURT/SyntaxFest 2023)*, 41–50. Washington, DC: Association for Computational Linguistics.

Boas, Hans C. 2013. Cognitive construction grammar. In Thomas Hoffmann & Graeme Trousdale (eds.), *The Oxford handbook of construction grammar*, 233–252. Oxford: Oxford University Press.

Boas, Hans C., Jaakko Leino & Benjamin Lyngfelt. 2024. Constructionist views on construction grammar. *Constructions and Frames* 16(2). 169–190.

Chen, Qiaoyun. 2025. Profiling analytic causative construction in Chinese: A multifactorial analysis of diachronic change. *Corpus Linguistics and Linguistic Theory* 21(1). 201–236.

Cheng, Winnie. 2011. *Exploring corpus linguistics: Language in action*. London & New York: Routledge.

Cohen, Jacob. 1988. *Statistical power analysis for the behavioral sciences*, 2nd edn. Hillsdale, NJ: Lawrence Erlbaum.

Desagulier, Guillaume. 2022. Changes in the midst of a construction network: A diachronic construction grammar approach to complex prepositions. *Cognitive Linguistics* 33(2). 339–386.

Fillmore, Charles J. 2006. Frame semantics. In Dirk Geeraerts (ed.), *Cognitive linguistics: Basic readings*, 373–400. Berlin & New York: Mouton de Gruyter.

Gilquin, Gaëtanelle. 2006. The verb slot in causative constructions: Finding the best fit. *Constructions* SV1-03. 1–46.

Goldberg, Adele E. 1995. *Constructions: A Construction grammar approach to argument structure*. Chicago: University of Chicago Press.

Gries, Stefan Th. & Anatol Stefanowitsch. 2003. Collostructions: Investigating the interaction of words and constructions. *International Journal of Corpus Linguistics* 8(2). 209–243.

Gries, Stefan Th. & Anatol Stefanowitsch. 2004. Extending collostructional analysis: A corpus-based perspective on alternations. *International Journal of Corpus Linguistics* 9(1). 97–129.

Hsiao, Huichen S. & Mahastuti Lestari. 2020. A collostructional analysis of ditransitive constructions in Mandarin. In Jia-Fei Hong, Yangsen Zhang & Pengyuan Liu (eds.), *Chinese lexical semantics: 20th workshop, CLSW 2019, Revised selected papers (lecture notes in computer science 12348)*, 37–51. Cham: Springer.

Kemmer, Suzanne & Arie Verhagen. 1994. The grammar of causatives and the conceptual structure of events. *Cognitive Linguistics* 5(2). 115–156.

Levin, Beth. 1993. *English verb classes and alternations: A preliminary investigation*. Chicago: University of Chicago Press.

Liesenfeld, Andreas, Meichun Liu & Chu-Ren Huang. 2022. Profiling the Chinese causative construction with *ràng*, *shǐ*, and *lìng* using frame semantic features. *Corpus Linguistics and Linguistic Theory* 18(2). 263–306.

Lin, Yen-Yu & Siaw-Fong Chung. 2016. A corpus-based study on the semantic prosody of challenge. *Taiwan Journal of TESOL* 13(2). 99–146.

Liu, Na & Fuyin Li. 2023. Mandarin analytic causative constructions with *shǐ* and *ràng*: A usage-based collostructional analysis. In Qi Su, Ge Xu & Xiaoyan Yang (eds.), *Chinese lexical semantics: CLSW 2022*, 539–552. Cham: Springer.

Partington, Alan. 1998. *Patterns and meanings: Using corpora for English language research and teaching*. Amsterdam & Philadelphia: John Benjamins.

Phoocharoensil, Supakorn. 2021. Semantic prosody and collocation: A corpus study of the near-synonyms persist and persevere. *Eurasian Journal of Applied Linguistics* 7(1). 240–258.

Pijpops, Dirk, Dirk Speelman, Freek Van de Velde & Stefan Grondelaers. 2021. Incorporating the multi-level nature of the construction into hypothesis testing. *Cognitive Linguistics* 32(3). 487–528.

R Core Team. 2023. *R: A language and environment for statistical computing*. Vienna: R Foundation for Statistical Computing. Available at: https://www.R-project.org/.

Shih, Ching-Fang, Mao-Chang Ku & Shu-Kai Hsieh. 2020. Lectal variation of the two Chinese causative auxiliaries. In *Proceedings of the 32nd Conference on Computational Linguistics and Speech Processing (ROCLING 2020)*, 163–177. Taipei: ACLCLP.

Sinclair, John. 1991. *Corpus, concordance, collocation*. Oxford: Oxford University Press.

Stubbs, Michael. 1995. Collocations and semantic profiles: On the cause of the trouble with quantitative methods. *Functions of Language* 2(1). 1–33.

Tian, Xiaoyu. 2025. Refining the network for the Chinese analytic causative construction: Insights from corpus analysis and psycholinguistic experimentation. *Corpus Linguistics and Linguistic Theory*. (advance online publication) https://www.degruyterbrill.com/document/doi/10.1515/cllt-2024-0071/html.

Tian, Xiaoyu, Weiwei Zhang & Speelman Dirk. 2022. Lectal variation in Chinese analytic causative constructions: What trees can and cannot tell us. In Dennis Tay & Molly Xie Pan (eds.), *Data analytics in cognitive linguistics: Methods and insights*, 137–168. Berlin & Boston: De Gruyter Mouton.

Verhagen, Arie & Suzanne Kemmer. 1997. Interaction and causation: Causative constructions in modern Standard Dutch. *Journal of Pragmatics* 27(1). 61–82.

Wang, Ying-Ting. 2011. A semantic, discourse, and pragmatic analysis of the Mandarin causative construction rang with pedagogical application [漢語致使結構之語義、篇章、語用分析及教學初探——以「讓」字句為例]. MA thesis. Taipei: National Taiwan Normal University.

Weng, Chuan-Hui. 2007. Causative, permissive, and yielding: The Mandarin Chinese verb *rang*. *Nanzan Linguistics* 1(2). 69–90.

Xiong, Xue-Liang & Xiao-Bo Liang. 2003. A prototypical study of the causative construction [致使結構的原型研究]. *Journal of Jiangxi Normal University (Philosophy and Social Sciences Edition)* 36(6). 106–110.

Yang, Jiang-Feng. 2016. A multi-dimensional study of periphrastic causative constructions in Mandarin Chinese [漢語迂回致使結構的多維度研究]. PhD dissertation. Hangzhou: Zhejiang University.

Zhang, Li-Li. 2005. The causativisation of Chinese shi-yi constructions [從使役到致使]. *Humanitas Taiwanica* 62. 119–152.

Zhang, Li-Li. 2006. The semantic development of passive meaning in Mandarin causative structure [漢語使役句表被動的語義發展]. *Language and Linguistics* 7(1). 139–174.

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